SAMUEL CHASSOT

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scholar.google.com/citations?user=yL85RAUAAAAJhl=froi=ao 0009-0000-9751-925 02 May 1996



EDUCATION

PhD in Computer Science
EPFL

Sep 2022 – Ongoing Lausanne, Switzerland

Master of Science and Bachelor of Science in Computer Science
EPFL

Sep 2016 – Feb 2022 Lausanne, Switzerland

RESEARCH EXPERIENCE

Verifying a Realistic Mutable Hash Table
LARA - EPFL

- Formal verification of the `LongMap` implementation of the Scala standard library with Stainless
- Discovery of a bug in the original implementation and proposition of a fix
- Release of the SMT queries generated by Stainless
- Published at IJCAR24

ASN.1 implementation in collaboration with ESA
LARA - EPFL

- Verification of an ASN.1 serialisation protocol implementation with Stainless
- Collective effort in my research group
- Collaboration with the European Space Agency
- Published at VMCAI25

A Verified Linear-Time Invertible Lexer, including Zipper-Based Regular Expression Matching and Memoization
LARA - EPFL

- Implementation of a linear-time lexer using regular expression
- Implementation and verification of a regular expression matching algorithm based on derivatives including a zipper-based optimization
- Verification of the correctness of the lexer including longest match semantics and invertibility in Stainless including memoization
- Reusable and fully verified memoization framework based on a verified mutable hash table

SVSHI - Secure and Verified Smart Home Infrastructure
Master thesis in DSLAB - EPFL

- Development of an open-source platform to write, compile, verify, and run Python applications on KNX infrastructures to bring formal verification into smart buildings

Scaling up an Unsupervised Image-to-Image Translation Framework from Basic to Complex Scenes
Research project in IVRL - EPFL

- Published in the American Journal of Computer Science and Technology 4

Development of a NIC driver in C#
Master semester project in DSLAB - EPFL

- Development of a NIC driver in C# based on a C version including performance evaluation showing the competitivity of high-level languages for drivers

Adding tooling support for Dotty Enum in Scalameta and Scalafmt
Bachelor semester project in LAMP - EPFL

- Modification of Scalameta and Scalafmt libraries to support the new Dotty Enum and development of a VS Code extension to format Scala code using *Scalafmt* and the LSP protocol

LANGUAGES

English
French
German

PROFESSIONAL EXPERIENCE

PhD Student: Teaching aspect
LARA EFPL

Sep 2022 – Ongoing Lausanne, Switzerland

- CS-214 - Software Construction
- CS-522 - Principles of Computer Systems
- CS-320 - Computer Language Processing
- 6 semesters as a Master student for courses such as software engineering

Applied Scientist Intern
AWS

May 2025 – Aug 2025 Austin, TX, USA

- Work on adding heap splitting reasoning to the Dafny verifier
- Work on heap splitting reasoning for Strata, an middle layer for verification tool offering intermediate representation, in Lean.

Research engineer
DSLAB EPFL

Apr 2022 – Sep 2022 Lausanne, Switzerland

- Supervision of a master thesis and some master semester projects related to the SVSHI project
- Research development in continuation to the SVSHI project

Software engineer
Swissquote

Mar 2022 – Apr 2022 Gland, Switzerland

Software engineer
BioT

Jul 2021 – Aug 2021 Lausanne, Switzerland

- Development of a solution based on Raspberry Pi and Python scripts including the management of the project
- Backend development (Kotlin + Vertx) and development of a triangulation program (Python + NumPy)

Software developer
Freelance

Jul 2018 – Aug 2022 Bulle, Switzerland

Software developer
Progin SA

Summers 2017 to 20 Bulle, Switzerland

- Various software development: mobile applications (iOS and Android), server-side DotNET applications, interoperability programs between systems (e.g., Microsoft Active Directory, Microsoft Dynamics NAV, and M-Files), Add-in for Microsoft Outlook, user-facing desktop applications with UI.